

ROADMAP.md

SynChain Product Roadmap

Version: 1.0 Status: Active Development

Vision

Build the world's most intelligent AI-powered Supply Chain Decision Intelligence Platform using Multi-Agent Systems and Digital Twin Technology.

Phase 0 — Research & Planning (Completed)

Objectives

- Identify real-world supply chain problems
- Define product vision
- Select architecture
- Design multi-agent workflow

Deliverables

- Product Requirements Document
- System Architecture
- Agent Design
- Database Design
- API Design

Status: Completed

Phase 1 — MVP Foundation

Duration: 2–3 Weeks

Goals

Build the initial working platform.

Frontend

- Landing Page
- Form Page
- Results Dashboard
- Responsive UI

Backend

- FastAPI Setup
- Database Setup
- Simulation Endpoint

Database

- User Tables
- Product Tables
- Simulation Tables

Deliverables

Users can create and run simulations.

Success Metric:

End-to-end simulation working.

Status: In Progress

Phase 2 — Multi-Agent System

Duration: 2 Weeks

Goals

Replace static logic with specialized agents.

Agents

Demand Agent

Forecast demand.

Inventory Agent

Optimize stock.

Logistics Agent

Select warehouse and route.

Risk Agent

Assess risks.

Recommendation Agent

Generate strategy.

Explanation Agent

Explain decisions.

Deliverables

Multi-agent pipeline operational.

Success Metric:

Recommendations generated through agent collaboration.

Phase 3 — Digital Twin Engine

Duration: 2 Weeks

Goals

Create a virtual representation of supply chain operations.

Components

- Inventory State
- Warehouse State
- Supplier State
- Transportation State
- Market State

Deliverables

Agents operate on virtual supply chain states.

Success Metric:

Digital Twin becomes the primary decision model.

Phase 4 — Scenario Simulation

Duration: 1–2 Weeks

Goals

Enable "What-If" analysis.

Scenarios

- Demand Surge
- Supplier Failure
- Transportation Delay
- Warehouse Closure
- Market Expansion

Deliverables

Interactive scenario simulation.

Success Metric:

Users can compare future outcomes.

Phase 5 — External Intelligence Layer

Duration: 2 Weeks

Goals

Make SynChain aware of external events.

Sources

- Tavily
- GDELT
- RSS Feeds

Intelligence Categories

- Manufacturing
- Logistics
- Retail
- Commodities
- Energy
- Technology

Deliverables

External signals influence recommendations.

Success Metric:

News events affect forecasts and risk analysis.

Phase 6 — Enterprise Platform

Duration: 3–4 Weeks

Features

- Authentication
- Role-Based Access
- Company Workspaces
- Team Collaboration
- Saved Simulations
- Reports

Deliverables

Enterprise-ready platform.

Success Metric:

Multiple organizations can use SynChain.

Phase 7 — Predictive Procurement

Features

- Supplier Ranking
- Purchase Timing
- Cost Optimization

Phase 8 — Autonomous Decision Support

Features

- Auto Recommendations
- Continuous Monitoring
- Automated Alerts

Long-Term Vision

SynChain evolves from a simulation platform into a real-time Supply Chain Operating System capable of assisting organizations in strategic, operational, and tactical decision-making.